

Method of Moment Estimation

Definition (Moment of a random variable)

Let X be a random variable and $c \in \mathbb{R}$ be a scalar. Then:

The k^{th} moment of $X =$

and

The k^{th} moment of $X(\text{about } c) =$

Definition (Sample moment)

Let X be a random variable, and $c \in \mathbb{R}$ a scalar.

The k^{th} sample moment of X =

and

The k^{th} sample moment of X (about c) =

Eg 1. Let's say $X_1, X_2, \dots, X_n$ are iid from $X \sim \text{Exp}(\theta)$. (These values may look like $x_1 = 3.21$, $x_2 = 5.11$, $x_3 = 4.33$, etc.) What is the MoM estimator of θ ?

Method of Moments Estimation

Eg 2. Let's say $X_1, X_2, \dots, X_n$ are iid samples from $X \sim \text{Pois}(\theta)$. (These values might look like $X_1 = 13$, $X_2 = 5$, $X_3 = 4$, etc.) What is the MoM estimator of θ ?

Eg 3. Let's say $X_1, X_2, \dots, X_n$ are iid samples from $X \sim N(\theta_1, \theta_2)$. (These values might look like $X_1 = -2.321$, $X_2 = 1.112$, $X_3 = -5.221$, etc.) What is the MoM estimator of the vector $\theta = (\theta_1, \theta_2)$ (θ_1 is the mean, and θ_2 is the variance)?

Eg 4. Let $X_1, \dots, X_n$ be a random sample from

$$f_{\theta}(x) = \begin{cases} (\theta + 1)x^{\theta} & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the method of moments estimator (MOM) for θ .

